

Python Practical Solutions

24 Programs on for/while Loops, break, continue, pass, Functions, Input Handling and Basic Problem Solving

1. First Prime Number Between 1 and 100

```
for num in range(2, 101):
    is_prime = True

    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break

    if is_prime:
        print("First prime number:", num)
        break
```

Output: First prime number: 2

2. Even Numbers from 1 to 50, Skip Multiples of 4

```
for num in range(1, 51):
    if num % 2 != 0:
        continue

    if num % 4 == 0:
        continue

    print(num)
```

Output: 2, 6, 10, 14, 18, 22, 26, 30, 34, 38, 42, 46, 50

3. Function to Print Even Numbers Using pass

```
def print_even_numbers(numbers):
    for num in numbers:
        if num % 2 == 0:
            print(num)
        else:
            pass

numbers = [10, 15, 20, 25, 30, 35]
print_even_numbers(numbers)
```

4. Multiplication Tables 1 to 10 — Break if Product > 50

```
for num in range(1, 11):
    for multiplier in range(1, 11):
        product = num * multiplier

        if product > 50:
            break

    print(f"{num} x {multiplier} = {product}")

print()
```

5. While Loop — Skip Multiples of 5

```
num = 1

while num <= 20:
    if num % 5 == 0:
        num += 1
        continue

    print(num)
    num += 1
```

6. Skip Numbers Divisible by 3 and Break at 25

```
for num in range(1, 31):
    if num == 25:
        break

    if num % 3 == 0:
        continue

    print(num)
```

7. Count Strings Containing 'a', Skip Empty Strings

```
def count_strings_with_a(words):
    count = 0

    for word in words:
        if word == "":
            continue

        if "a" in word.lower():
            count += 1

    return count

words = ["apple", "", "banana", "cat", "dog", ""]
result = count_strings_with_a(words)

print("Count:", result)
```

Output: Count: 3

8. Use pass for Numbers Less Than 5

```
for num in range(1, 11):
    if num < 5:
        pass
    else:
        print(num)
```

Output: 5, 6, 7, 8, 9, 10

9. Sum Odd Numbers from 1 to 100 Using continue

```
total = 0

for num in range(1, 101):
    if num % 2 == 0:
```

```

        continue

    total += num

print("Sum of odd numbers:", total)

```

Output: Sum of odd numbers: 2500

10. Print 5x5 Grid, Skip Diagonal Cells

```

for row in range(5):
    for col in range(5):
        if row == col:
            continue

        print("*", end=" ")

    print()

```

Diagonal cells are skipped because row == col.

11. Ask for Numbers Until Negative Number

```

while True:
    num = int(input("Enter a number: "))

    if num < 0:
        print("Negative number entered. Program stopped.")
        break

    print("You entered:", num)

```

12. User Inputs Until 'done', Skip Non-Numbers

```

while True:
    user_input = input("Enter a number or 'done': ")

    if user_input.lower() == "done":
        break

    try:
        number = float(user_input)
        print("Number:", number)
    except ValueError:
        continue

```

13. Right Triangle Pattern — Break if Stars Exceed 10

```

for rows in range(1, 11):
    if rows > 10:
        break

    print("*" * rows)

```

14. Factorial Using Loop — Break if Number Is Negative

```

num = int(input("Enter a number: "))

if num < 0:
    print("Factorial is not defined for negative numbers.")

```

```

else:
    factorial = 1

    for i in range(1, num + 1):
        factorial *= i

    print("Factorial:", factorial)

```

15. Cumulative Sum Until User Enters 0

```

total = 0

while True:
    num = int(input("Enter a number: "))

    if num == 0:
        break

    total += num

print("Cumulative sum:", total)

```

16. Print 1 to 100, Skip Numbers Containing Digit '5'

```

for num in range(1, 101):
    if "5" in str(num):
        continue

    print(num)

```

17. Remove Zeros Using continue

```

def remove_zeros(numbers):
    result = []

    for num in numbers:
        if num == 0:
            continue

        result.append(num)

    return result

numbers = [10, 0, 20, 0, 30, 40, 0]
print(remove_zeros(numbers))

```

Output: [10, 20, 30, 40]

18. Count Vowels Using pass for Consonants

```

def count_vowels(text):
    vowels = "aeiou"
    count = 0

    for char in text.lower():
        if char in vowels:
            count += 1
        else:
            pass

    return count

```

```
text = input("Enter a string: ")
print("Number of vowels:", count_vowels(text))
```

19. Check Duplicates — Skip Already Seen Numbers

```
numbers = [10, 20, 10, 30, 20, 40, 50, 30]

seen = set()
duplicates = []

for num in numbers:
    if num in seen:
        duplicates.append(num)
        continue

    seen.add(num)

print("Duplicate numbers:", duplicates)
```

Output: Duplicate numbers: [10, 20, 30]

20. Guessing Game — Random Number 1 to 100

```
import random

secret_number = random.randint(1, 100)

while True:
    user_input = input("Guess the number or type 'quit': ")

    if user_input.lower() == "quit":
        print("Game ended.")
        break

    try:
        guess = int(user_input)
    except ValueError:
        print("Please enter a valid number.")
        continue

    if guess < secret_number:
        print("Too low!")
    elif guess > secret_number:
        print("Too high!")
    else:
        print("Congratulations! You guessed it.")
        break
```

21. Sum Non-Negative Integers Until Negative

```
total = 0

while True:
    num = int(input("Enter an integer: "))

    if num < 0:
        break

    total += num

print("Total:", total)
```

22. Print Prime Factors of a Number

```
def is_prime(num):
    if num < 2:
        return False

    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            return False

    return True

def prime_factors(number):
    for num in range(2, number + 1):
        if number % num != 0:
            continue

        if is_prime(num):
            print(num)

number = int(input("Enter a number: "))
prime_factors(number)
```

Example: For 60, output is 2, 3, 5.

23. Even Numbers 1 to 50 in Reverse Using continue

```
for num in range(50, 0, -1):
    if num % 2 != 0:
        continue

    print(num)
```

Output starts: 50, 48, 46, 44, ... , 4, 2

24. Ask for Input Until 'exit', Skip Empty Inputs

```
while True:
    user_input = input("Enter something: ")

    if user_input.lower() == "exit":
        print("Program ended.")
        break

    if user_input.strip() == "":
        continue

    print("You entered:", user_input)
```

Empty inputs are skipped; the program exits when the user types exit.

Quick Concept Reference

break — Immediately terminates the current loop.

continue — Skips the current iteration and moves to the next iteration.

pass — Does nothing; commonly used as a placeholder.

for — Used to iterate over a sequence or range.

while — Repeats while a condition remains true.

function — Reusable block of code defined with `def`.